UNIVERSITY OF GENEVA

Faculty of Science

My Course

Digital Forensics
14x065

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1 Introduction

Humans have always tried to capture the present moment, both in painting and in photography. However, the photos taken are not always of good quality, that's why some computational methods, such as image filtering, have been developped to increase the quality of a given picture.

In this project, we will examine some basic image filtering methods and some basic way to reconstruct damaged pictures.

2 Methodology

The image filtering works in both spatial and Fourier domain.

The basics are the following:

- A filter is created to apply some operation on the image.

For instance, a laplacian filter gives the gradient of the image.

- The filter is applied to the image in spatial or Fourier domain.
  - In spatial domain, the convolution between the image and the filter is performed to apply the filter to each pixel of the image.
  - In Fourier domain, a simple multiplication between the image and the filter applies the filter to the image. Original image filter

3 Implementation

4 Results

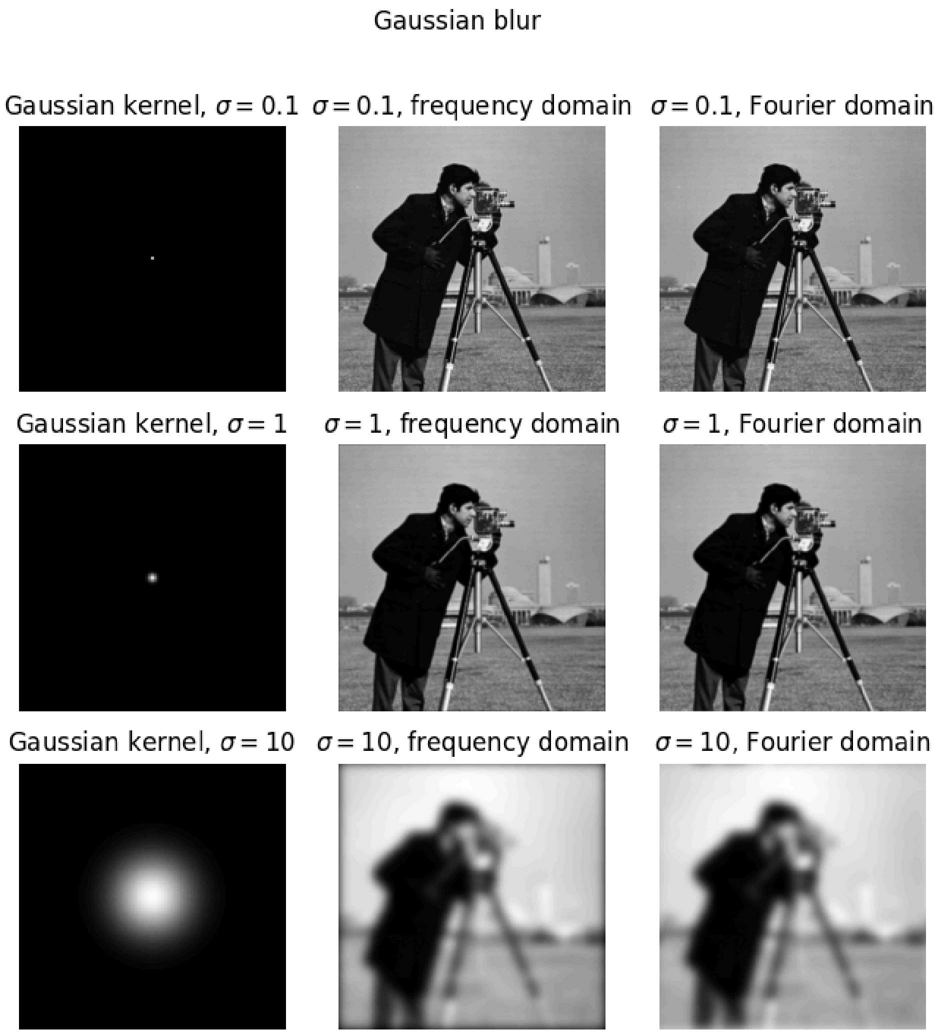


Figure 1: Low-Pass filter: Gaussian Blur

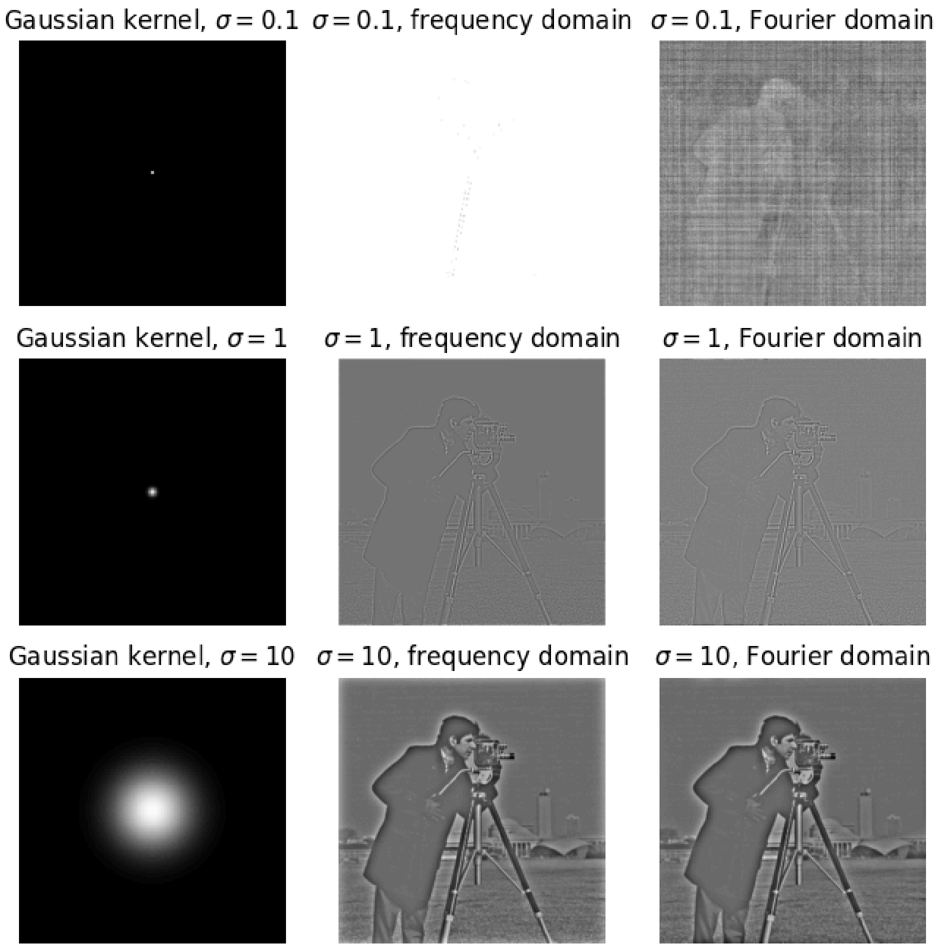
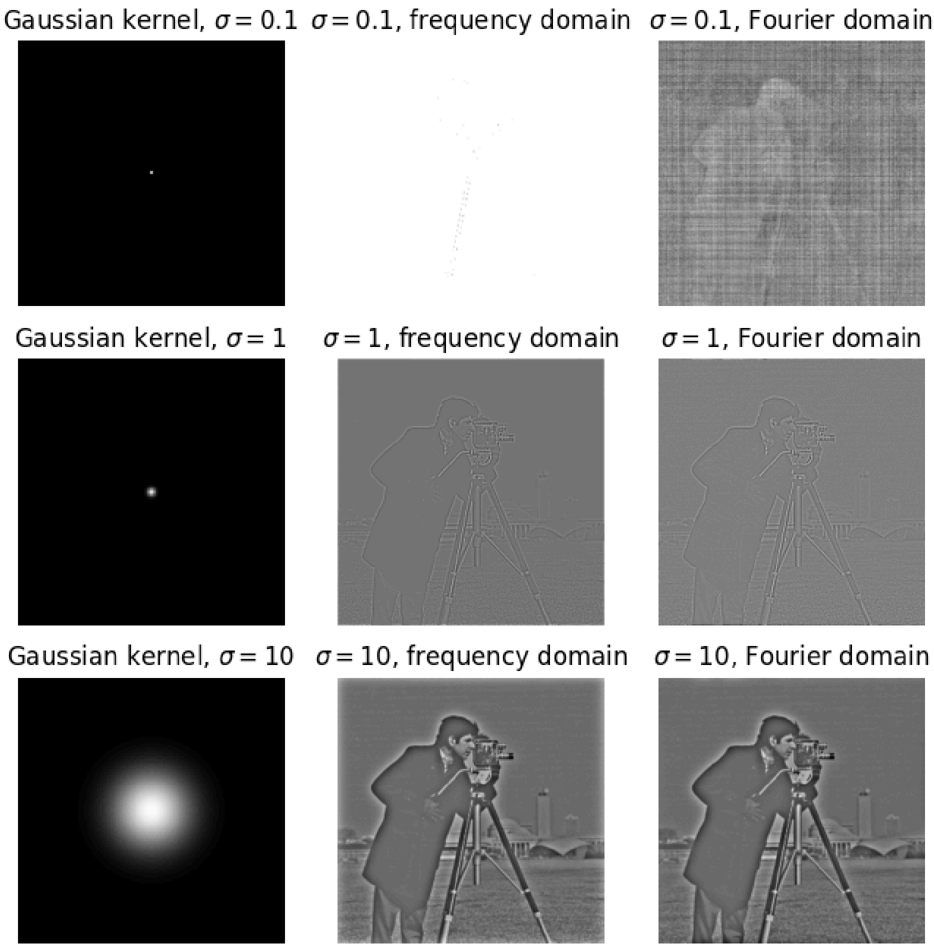
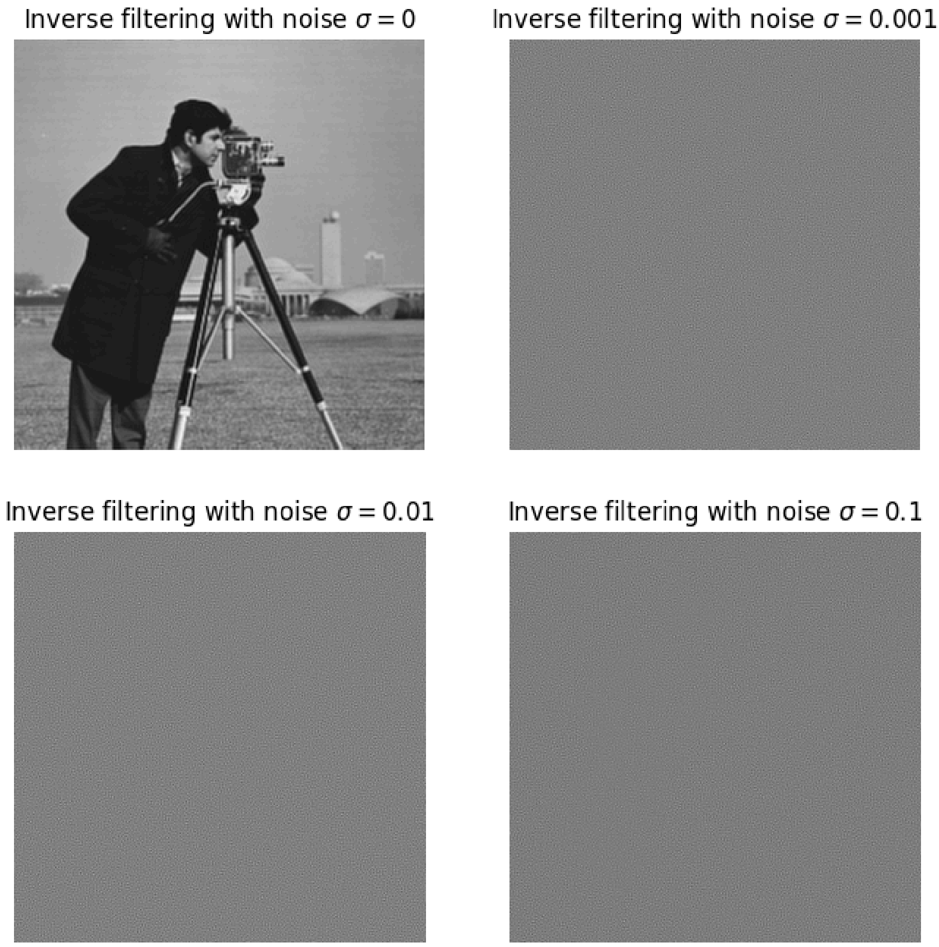
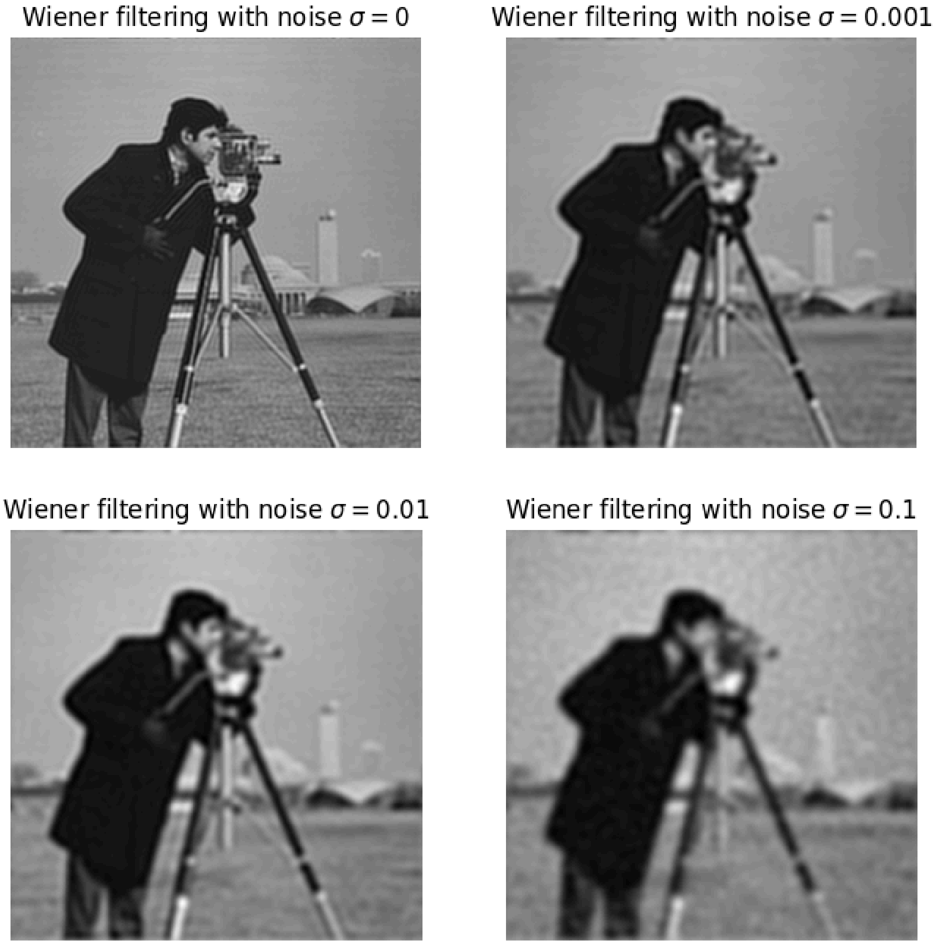


Figure 2: High-Pass filter: Unsharp Mask


Gaussian kernel, $\sigma = 0.1$
$\sigma = 0.1$, frequency domain
$\sigma = 0.1$, Fourier domain
Gaussian kernel, $\sigma = 1$
$\sigma = 1$, frequency domain
$\sigma = 1$, Fourier domain
Gaussian kernel, $\sigma = 10$
$\sigma = 10$, frequency domain
$\sigma = 10$, Fourier domain


Inverse filtering with noise $\sigma = 0$
Inverse filtering with noise $\sigma = 0.001$
Inverse filtering with noise $\sigma = 0.01$
Inverse filtering with noise $\sigma = 0.1$



5 Discussion

Blabla Blabla

6 Conclusion